

1. List all 6 phases of SDLC in order with one line about each.

Phase	Description
1. Requirement Analysis	Gather and analyze the needs of users and stakeholders.
2. Design	Create the system architecture and design specifications.
3. Development (Coding)	Developers write the code for the software.
4. Testing	The software is tested to find and fix defects.
5. Deployment	The software is released and made available to users.
6. Maintenance	The software is updated, improved, and supported after release.

2. Write 3 differences between Agile and Waterfall models.

Agile	Waterfall
Development is done in small iterations.	Development follows a fixed sequence of phases.
Changes can be made easily during the project.	Changes are difficult once a phase is completed.
Customer feedback is collected regularly.	Customer feedback is usually received at the end.

3. Name the 3 main Scrum roles and write one line about the responsibility of each.

Scrum Role	Responsibility
Scrum Master	Facilitates the Scrum process and removes obstacles for the team.
Product Owner	Defines requirements and prioritizes the product backlog.

Scrum Role	Responsibility
Development Team	Designs, develops, and tests the product increment.

4. What is a Sprint in Scrum? Mention its typical duration.

A **Sprint** is a fixed period during which a Scrum team completes a set of planned tasks and delivers a working product increment.

Typical Duration: 1 to 4 weeks (commonly 2 weeks).

5. What do these Jira board columns mean: To Do, In Progress, Done, Backlog?

Column	Meaning
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Backlog	List of tasks or requirements that are planned for future work.
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To Do	Tasks that are ready to be started.
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In Progress	Tasks that are currently being worked on.
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Done	Tasks that have been completed successfully.
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